CNC-204 Milling Machine

Technical Maintenance & Repair Manual

1. Overview

The CNC-204 is a 3-axis precision milling machine used for machining metal and composite parts. This manual covers routine maintenance, common fault diagnosis, and step-by-step repair procedures for field technicians. Always isolate power and follow lockout/tagout (LOTO) procedures before performing any maintenance.

2. Fault Code E-108: Spindle Overheating

Error E-108 is triggered when the spindle temperature sensor reports a reading above 75 degrees C for more than 30 continuous seconds. Overheating is most commonly caused by worn spindle bearings, insufficient coolant flow, or a degraded coolant pump. Left unaddressed, this can cause permanent spindle damage.

2.1 Diagnostic Steps

Step 1: Power down the machine and engage lockout/tagout.

Step 2: Inspect the coolant reservoir level; refill if below the minimum line.

Step 3: Check the coolant pump (Part SKU: CLP-0876) for flow output; replace if flow is below 2 L/min.

Step 4: Remove the spindle housing cover and inspect the spindle bearing set (Part SKU: SPB-2201) for pitting, discoloration, or excessive play.

Step 5: If bearing play exceeds 0.05mm radial clearance, replace the spindle bearing set.

Step 6: Reassemble, refill coolant, and run a 10-minute idle test while monitoring spindle temperature.

2.2 Torque Specifications

Component	Fastener	Torque (Nm)
Spindle housing cover	M6 bolt	9.5
Bearing retainer plate	M5 bolt	6.0
Coolant pump mount	M8 bolt	18.0
Drive belt tensioner	M6 bolt	9.5

3. Fault Code E-212: Drive Belt Slippage

Error E-212 indicates the X-axis drive belt encoder has detected slippage greater than 2 percent between commanded and actual position. This is usually caused by a worn or under-tensioned drive belt (Part SKU: DBT-1145).

3.1 Repair Procedure

Step 1: Power down and lockout/tagout the machine.

Step 2: Remove the axis cover panel (4x M4 screws).

Step 3: Inspect the drive belt for cracking, glazing, or fraying.

Step 4: If damaged, replace with Drive Belt (Type A), SKU DBT-1145.

Step 5: Set belt tension using a tension gauge to 45-50 N; adjust via the tensioner bolt to 9.5 Nm torque.

Step 6: Reinstall the axis cover and run the axis homing cycle to confirm the fault has cleared.

4. Fault Code E-330: Servo Encoder Fault

Error E-330 signals a loss of signal from the servo motor encoder (Part SKU: SME-5521), typically caused by a damaged encoder cable or a failed encoder unit. This fault requires immediate machine shutdown to prevent uncontrolled axis motion.

4.1 Repair Procedure

Step 1: Power down and lockout/tagout the machine immediately.

Step 2: Disconnect and visually inspect the encoder cable for pinching or fraying.

Step 3: If the cable is intact, test encoder output using a multimeter at the terminal block.

Step 4: If no signal is present, replace the Servo Motor Encoder, SKU SME-5521.

Step 5: Recalibrate axis zero position after replacement using the controller's homing wizard.

Step 6: Verify by jogging the axis and confirming smooth position feedback with no fault codes.

4.2 Safety Notice

Only qualified technicians should perform electrical repairs. Always verify zero-energy state with a voltage tester before touching any electrical components.